CSI 2101: Discrete Structures

3. INDUCTION

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Theorem: $\forall n \geq 1\colon\ 1 + 2 + \ldots + n = n(n+1)/2.$

n	$1+2+\ldots+n$	$n(n+1)/2$
1	1	$1 \cdot 2/2 = 1$
2	$1+2=3$	$2 \cdot 3/2 = 3$
3	$1+2+3=6$	$3 \cdot 4/2 = 6$
4	$1+2+3+4=10$	$4 \cdot 5/2 = 10$

USE $P(n)$ WHEN PROVING $P(n+1)$.

proof by induction

$$\frac{P(1) \qquad P(1) \to P(2) \qquad P(2) \to P(3) \qquad P(3) \to P(4)}{P(1) \text{ AND } P(2) \text{ AND } P(3) \text{ AND } P(4)}$$

BASE CASE

INDUCTIVE STEP

INDUCTIVE HYPOTHESIS

$$\frac{P(1) \qquad \forall n\colon P(n) \to P(n+1)}{\forall n\colon P(n)}$$

Theorem: $1 + 2 + \ldots + n = n(n+1)/2.$

Proof: BASE CASE: $1 = \frac{1 \cdot 2}{2}$.

INDUCTIVE STEP:

ASSUME $1 + \ldots + n = \frac{n(n+1)}{2}$ (IND. HYPOTHESIS)

THEN $1 + \ldots + (n+1) = \frac{n(n+1)}{2} + (n+1)$

$$= \left(\frac{n}{2} + 1\right)(n+1)$$

$$= \frac{(n+1)(n+2)}{2}.$$

$\square$

Theorem: $\forall n \geq 4: \; n! > 2^n.$

$n! = 1 \cdot 2 \cdot \ldots \cdot n.$

Proof BY INDUCTION ON n.

BASE CASE: $4! = 24 > 16 = 2^4.$

INDUCTIVE STEP: ASSUME $n! > 2^n.$

$$(n+1)! = n!\,(n+1) \underset{\substack{\uparrow \\ \text{INDUCTIVE} \\ \text{HYPOTHESIS}}}{>} 2^n \cdot (n+1) \underset{\substack{\uparrow \\ \text{CAN USE} \\ n \geq 4 \\ \text{SO } n+1 > 2.}}{>} 2^n \cdot 2 = 2^{n+1}. \quad \square$$

Theorem: For odd $n \geq 1$, n^2 is of the form $8k + 1$.

$1 \quad 3 \quad 5 \quad 7 \quad \cdots n \quad n+2 \quad \cdots$

BASE CASE $n=1$: $1^2 = 1 = 8 \cdot 0 + 1$. ✓

INDUCTIVE STEP: ASSUME $n^2 = 8k+1$.

THEN $(n+2)^2 = n^2 + 4n + 4$

$$= n^2 + 4\underbrace{(n+1)}_{\text{EVEN}:\,=2t}$$

$$= (8k+1) + 4 \cdot 2t$$

$$= 8(k+t) + 1. \qquad \square$$

Theorem: $\forall n \geq 1$: 6 divides $n^3 - n$.

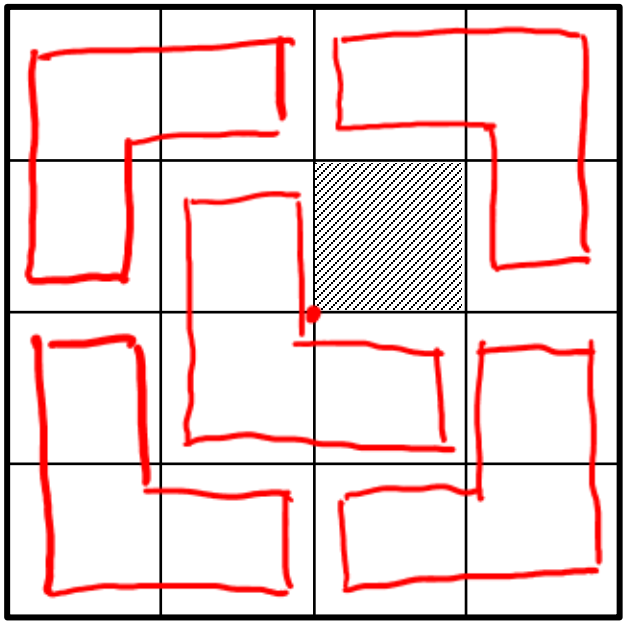
BASE CASE: $n=1$: $1^3 - 1 = 0 = 6 \cdot 0$. ✓

INDUCTIVE STEP: ASSUME 6 DIVIDES $n^3 - n$

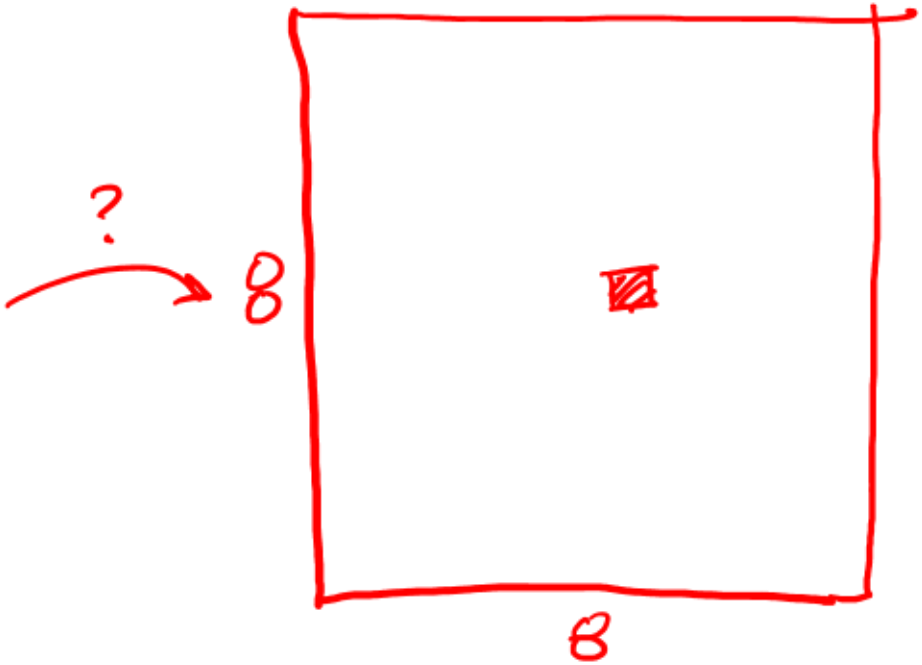
$$
\begin{aligned}
(n+1)^3 - (n+1) &= (n^3 + 3n^2 + 3n + 1) - (n+1) \\
&= n^3 + 3n^2 + 2n \\
&= (n^3 - n) + 3n^2 + 3n \\
&= \underbrace{(n^3 - n)}_{\substack{= 6k \\ \text{BY IND. HYP.}}} + 3\underbrace{n(n+1)}_{\text{EVEN} = 2t} \\
&= 6k + 3 \cdot 2t \\
&= 6(k+t).
\end{aligned}
$$

$\square$

A $2^n \times 2^n$ grid has one of its central squares removed. Can you fill it with L-shaped tiles?

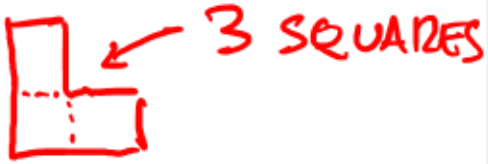


n = 2



n = 3

INADEQUATE PROOF:



BASE CASE $n=1$: $2\times 2-1 = 4-1 = 3$ ✓

INDUCTIVE STEP:

ASSUME $2^n \times 2^n - 1 = 3k$ FOR SOME k

THEN

$$
\begin{aligned}
2^{n+1}\times 2^{n+1}-1 &= 4\cdot 2^n\times 2^n - 1 \\
&= 3\cdot 2^n\times 2^n + (2^n\times 2^n - 1) \\
&= 3(2^n\times 2^n + k). \qquad \square
\end{aligned}
$$

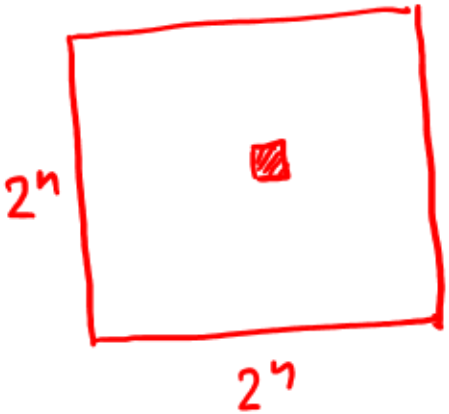
Theorem: A $2^n \times 2^n$ grid with a central square removed can be filled with L-shaped tiles.

BASE CASE $n=1$.

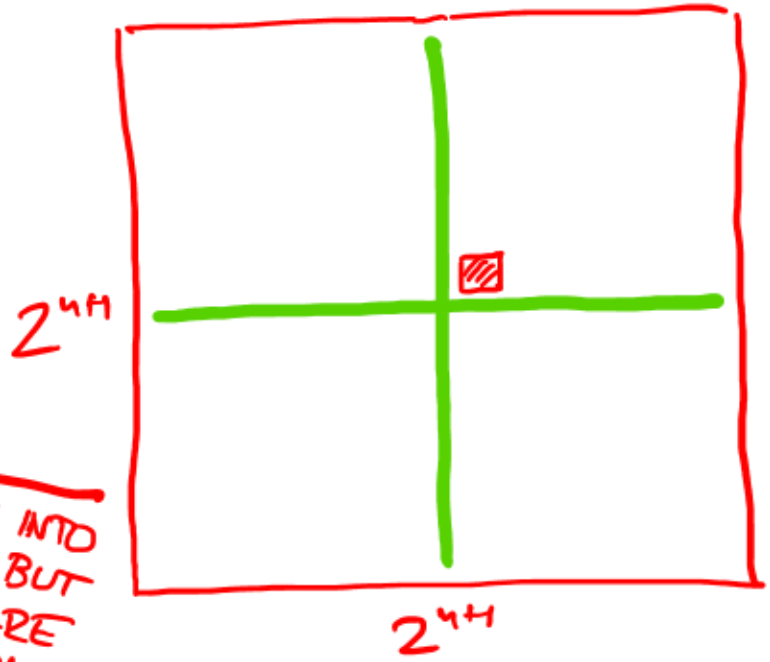
✓

INDUCTIVE STEP.

ASSUME CAN FILL



HOW TO DO

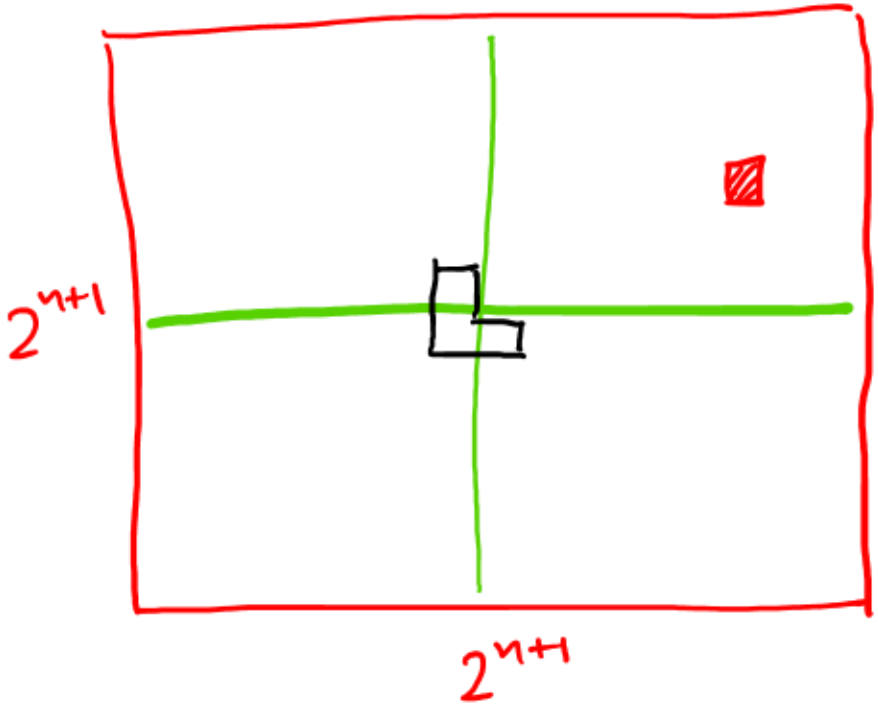


CAN SPLIT INTO FOUR $2^n \times 2^n$, BUT MISSING SQUARE IS NOT CENTRAL! (OR NOT MISSING)

Theorem: A $2^n \times 2^n$ grid with ~~a central~~ any one square removed can be filled with L-shaped tiles.

BASE CASE IS THE SAME

INDUCTIVE STEP



CAN ALWAYS ADD L TILE IN MIDDLE THAT REMOVING ITS SQUARES GIVES FOUR INSTANCES OF SIZE $2^n \times 2^n$ EACH WITH ONE SQUARE REMOVED. $\square$

ASSUMPTION CONCLUSION

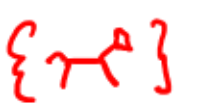
$$\frac{P(1) \qquad \forall n\colon P(n) \to P(n+1)}{\forall n\colon P(n)}$$

STRENGTHENING P(n) MAY HELP BECAUSE IT ALLOWS US TO ASSUME MORE!

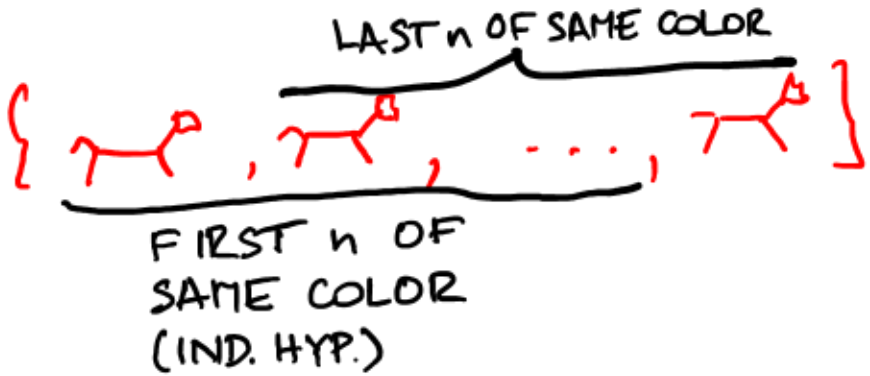
Theorem: All horses have the same colour.

<u>Proof.</u> BY INDUCTION ON THE NUMBER n OF HORSES IN THE SET.

($P(n)$: ALL SETS OF n HORSES CONSIST OF SAME-COLOR HORSES)

BASE CASE $n=1$:  ✓

INDUCTIVE STEP. ASSUME TRUE FOR n.
TAKE ANY SET OF $n+1$ HORSES.

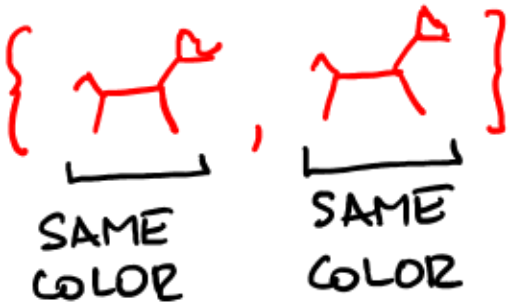


→ ALL OF SAME COLOR!

???

$P(1)$ $\quad$ $\boxed{P(1) \rightarrow P(2)}$ $\quad$ $P(2) \rightarrow P(3)$ $\quad$ $P(3) \rightarrow P(4)$

T $\quad$ F $\quad$ T $\quad$ T



CANNOT CONCLUDE <u>BOTH</u> HAVE THE SAME COLOR.

You have a stack of \$3 and \$5 stamps. Which dollar amounts can you make?

3 ✓

4 ✗

5 ✓

6 = 3+3 ✓

7 ✗

8 = 5+3

9 = 3+3+3

10 = 5+5

11 = 5+3+3

12 = 3+3+3+3

13 = 5+5+3

⋮

strong induction

$$\frac{P(1) \qquad \forall n: P(1) \text{ AND} \cdots \text{AND } P(n-1) \rightarrow P(n)}{\forall n: P(n)}$$

<u>Theorem.</u> EVERY $n \geq 8$ IS OF THE FORM $3a+5b$ FOR INTEGER $a, b \geq 0$.

<u>Proof.</u> BASE CASE: $8 = 5+3$

INDUCTIVE STEP: ASSUME TRUE FOR $8, 9, \ldots, n-1$.

$n = 9 = 3 \cdot 3$ ✓

$n = 10 = 5 \cdot 2$ ✓

$n \geq 11 \longrightarrow n-3 = 3a+5b$ (IND. HYP) $\rightarrow n = 3(a+1)+5b$ $\square$

The Fibonacci numbers are

$$\cancel{1} \quad 1 \quad 2 \quad 3 \quad 5 \quad 8 \quad 13 \quad 21 \quad 34$$

$$F_0 \quad F_1 \quad F_2 \quad F_3 \quad F_4 \quad F_5 \quad \cdots$$

$$F_1 = 1 \qquad F_2 = 2 \qquad F_{n+1} = F_n + F_{n-1}$$

Which numbers can be written as a sum of distinct Fibonacci numbers?

$4 = 3 + 1$

$6 = 5 + 1 = 3 + 2 + 1$

$7 = 5 + 2$

$9 = 8 + 1 = 5 + 3 + 1$

$10 = 8 + 2 = 5 + 3 + 2$

$\vdots$

$$(\text{REP. OF } m) = \underbrace{(\text{REP. OF } m-1) + 1}_{\text{MIGHT NOT BE DISTINCT!}}$$

Theorem. EVERY $m \geq 1$ CAN BE REPRESENTED AS A SUM OF DISTINCT FIBONACCIS.

Proof. BY STRONG INDUCTION ON m.

$m = 1$ ✓

ASSUME TRUE FOR 1 UP TO $m-1$.

CASE 1. $m = F_n$ FOR SOME n. ✓

CASE 2. $F_n < m < F_{n+1}$ FOR SOME n.

$$(\text{REP. OF } m) = F_n + \underbrace{(\text{REP. OF } m - F_n)}_{\text{ALL DISTINCT BY INDUCTIVE HYPOTHESIS}}$$

$m - F_n < F_{n+1} - F_n = F_{n-1} < F_n$ SO F_n IS ALSO DISTINCT FROM ALL OTHER NUMBERS IN REPRESENTATION. $\square$

ALGORITHM: GIVEN m

- IF m IS A FIBONACCI OUTPUT IT
- ELSE LET F_n BE LARGEST FIB. $< m$
- OUTPUT $F_n + ($REPRESENTATION OF $m - F_n)$.

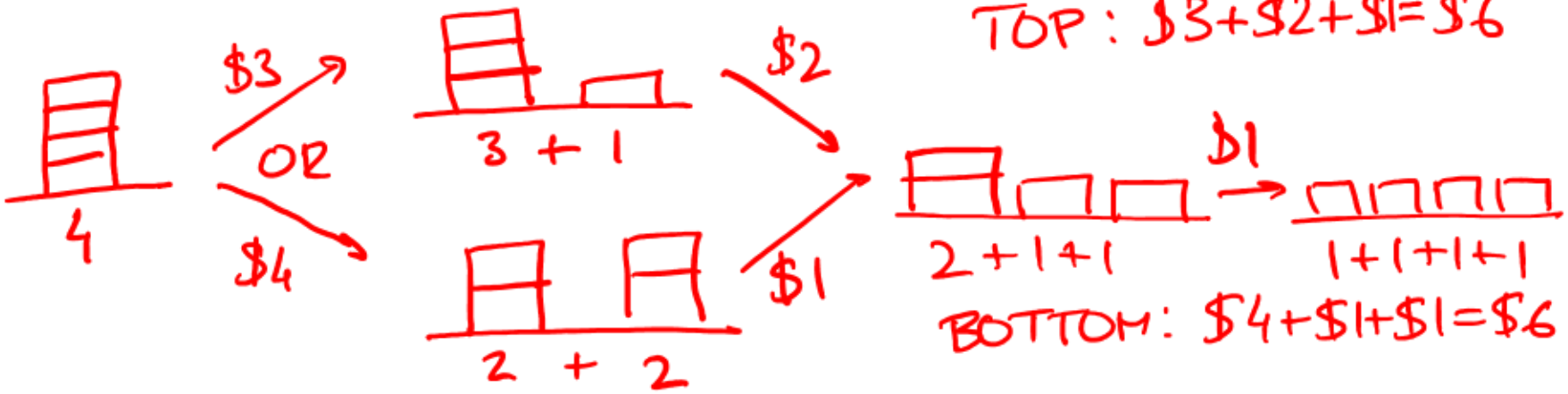
$$1 \quad 2 \quad \underset{\substack{\uparrow \\ 4}}{3} \quad 5 \quad 8 \quad \underset{\substack{\uparrow \\ 12}}{13} \quad 21 \quad 34 \quad \underset{\substack{\uparrow \\ 46}}{55}$$

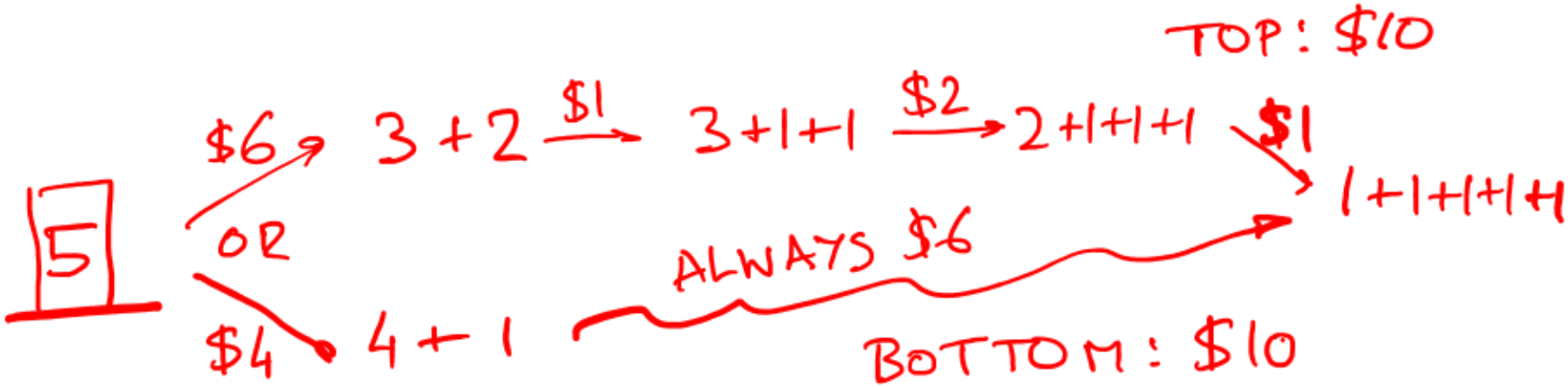
<u>Ex.</u> $m = 46 = 34 + 12$

$= 34 + 8 + 4$

$= 34 + 8 + 3 + 1$

You have a stack of ~~10~~ n boxes. Each time you split a stack you pay the product of heights. What is the minimum price to unstack?

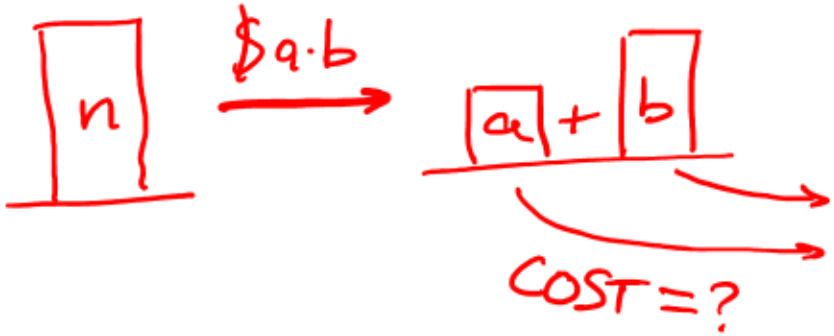




Conjecture. ALL UNSTACKING STRATEGIES FOR n BOXES COST THE SAME.

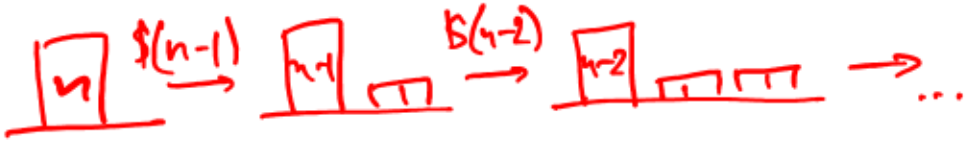
Proof. $n=1$: ONLY ONE STRATEGY. ✓

ASSUME TRUE UP TO $n-1$ BOXES.



???

COST FOR n BOXES SHOULD BE

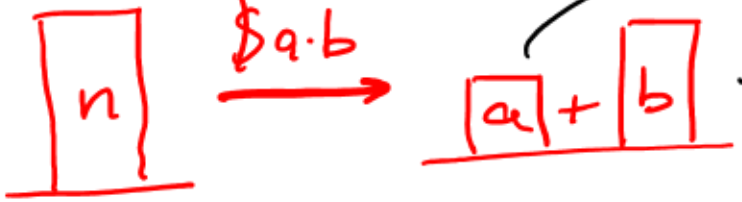


$$(n-1)+(n-2)+\dots+1=\frac{(n-1)n}{2}$$

Theorem. ALL UNSTACKING STRATEGIES FOR n BOXES COST ~~THE SAME~~. $\frac{(n-1)n}{2}$ DOLLARS.

Proof. $n=1$: COST $= 0 = \frac{(1-1)\cdot 1}{2}$ ✓

ASSUME TRUE UP TO $n-1$ BOXES.



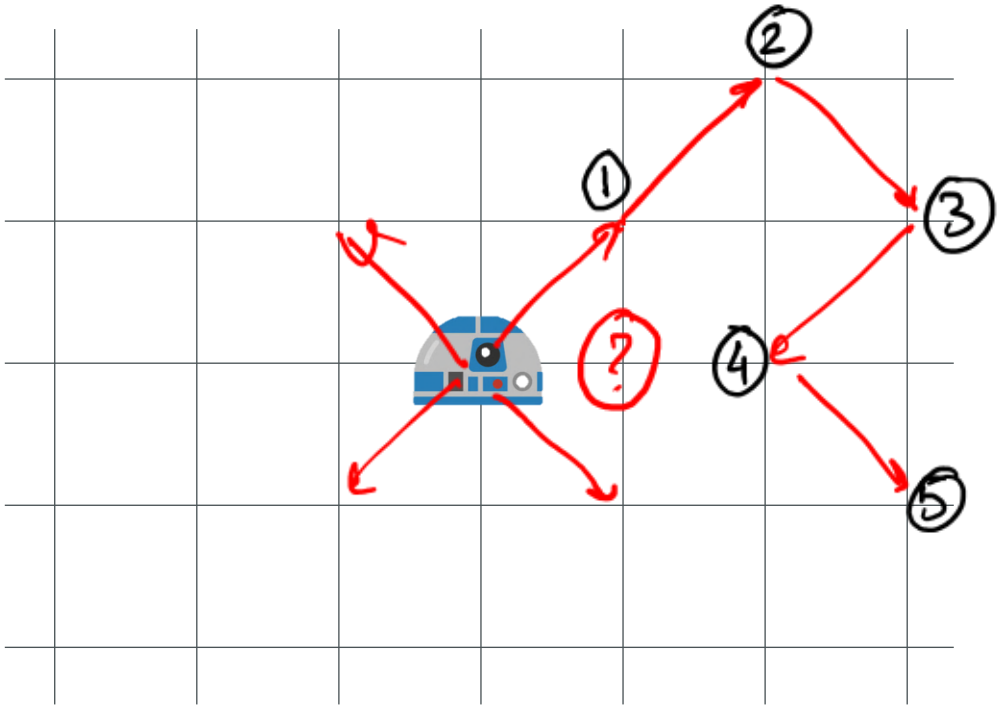
$$\left.\begin{array}{l} \text{COST} = \frac{(a-1)a}{2} \\ \text{COST} = \frac{(b-1)b}{2} \end{array}\right\} \text{INDUCTIVE HYPOTHESIS}$$

$$\text{COST FOR } n = ab + \frac{(a-1)a}{2} + \frac{(b-1)b}{2}$$

$$= \frac{2ab + a^2 - a + b^2 - b}{2}$$

$$= \frac{(a+b)^2 - (a+b)}{2} = \frac{(n-1)n}{2}. \qquad \square$$

A robot walks on the plane. Every step is 1 unit up/down and 1 unit left/right. Can it end up 1 unit to the right from where is started?



state machines

- set of states S
- start state $s_0 \in S$
- transition predicate $q \longrightarrow r$

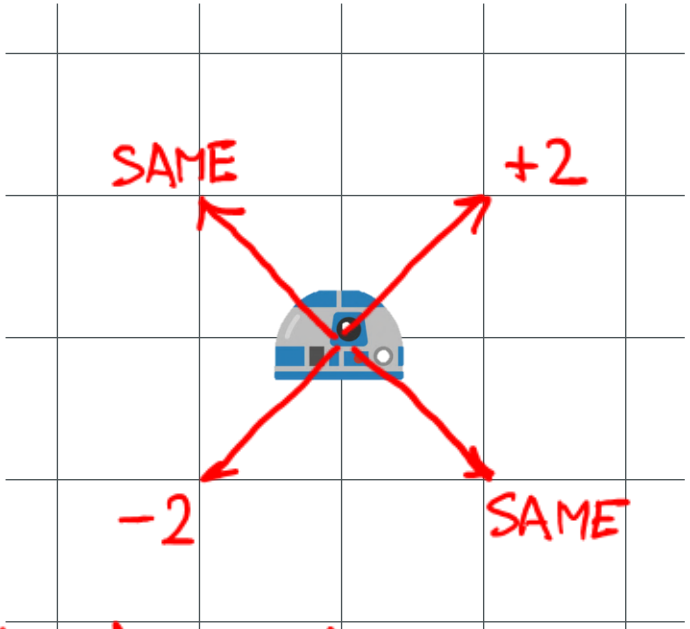
$$S = \{(0,0), (-1,0), (0,1), (3,5), \ldots\}$$
$$= \{(x,y) : x, y \text{ ARE INTEGERS}\}$$

$$s_0 = (0,0)$$

$$(x,y) \to \begin{cases} (x-1, y-1) \\ (x-1, y+1) \\ (x+1, y-1) \\ (x+1, y+1) \end{cases}$$

$P(x,y)$ = "$x+y$ IS EVEN."

$P(0,0)$ = "0 EVEN" T



ASSUME $x+y$ EVEN

$(x,y) \to (x-1,y-1)$ $\quad (x-1)+(y-1) = x+y-2$ EVEN

$(x,y) \to (x-1,y+1)$ $\quad (x-1)+(y+1) = x+y$ EVEN

$(x,y) \to (x+1,y-1)$ $\quad (x+1)+(y-1) = x+y$ EVEN

$(x,y) \to (x+1,y+1)$ $\quad (x+1)+(y+1) = x+y+2$ EVEN

SO $x+y$ MUST STAY EVEN AT ALL TIMES

$0+1$ IS ODD $\longrightarrow$ $(0,1)$ UNREACHABLE

invariants

An invariant is a predicate that is true for all reachable states

If P holds for the start state and is preserved by transitions, P is an invariant

P(s) IS FALSE → s IS UNREACHABLE

INVERSION

1	6	3
7	4	
5	2	8

INVERSIONS = 63, 64, 65, 62, 32, 74, 75, 72, 42, 52

1	2	3
4	5	6
8	7	

$\xrightarrow{?}$

1	2	3
4	5	6
7	8	

1 INVERSION

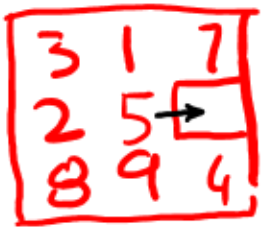
NO INVERSIONS

INVARIANT: NUMBER OF INVERSIONS IS ODD.

START STATE: 1 INVERSION

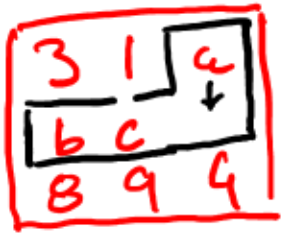
TRANSITIONS:

HORIZONTAL



NO CHANGE
STAYS ODD

VERTICAL



AFFECTS
ab, ac

CHANGES STATUS OF ab, ac SO NUMBER OF INVERSIONS CHANGES BY +2, 0, OR −2.

1	2	3
4	5	6
8	7	

$\xrightarrow{?}$

2	1	3
4	5	6
7	8	

1 INVERSION

1 INVERSION

Invariant(x) IS FALSE → x IS <u>NOT</u> REACHABLE

Invariant(x) IS TRUE → WE DON'T KNOW!

Ex.

"n IS EVEN" IS AN INVARIANT

UNREACHABLE BY INVARIANT

INVARIANT HOLDS BUT UNREACHABLE